

TASK 4 REPORT

Incident Response Simulation

Candidate Details

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1. Project Title

Incident Response Simulation

2. Objective

The objective of this task is to simulate a cybersecurity incident, analyze suspicious Windows security logs, identify Indicators of Compromise (IOCs), document response procedures, containment actions, and prepare an incident report suitable for security documentation.

3. Introduction

Incident Response (IR) is a structured process used to identify, investigate, contain, eradicate, and recover from cybersecurity incidents. Organizations rely on Windows Event Logs and Security Information and Event Management (SIEM) platforms to detect abnormal activities such as unauthorized logins, malware execution, privilege escalation, and suspicious network communication. A well-documented response minimizes damage and restores normal operations quickly.

4. Theory

Windows Event Viewer stores Security, System and Application logs. Security Event IDs such as 4624 (successful logon), 4625 (failed logon), 4688 (process creation), and 5156 (network connection) provide valuable evidence during investigations. Indicators of Compromise (IOCs) include repeated failed logins, unexpected PowerShell execution, unknown processes, suspicious outbound traffic, and abnormal user behavior. Security analysts correlate these events to determine the scope and severity of an incident.

5. Tools Used

- Windows Event Viewer
- Sample SIEM Logs
- Microsoft Defender (reference)
- Microsoft Word
- Windows Operating System

6. Simulated Incident Scenario

A workstation generated multiple failed login attempts followed by a successful login from the same account. Shortly afterward, PowerShell executed unexpectedly and outbound network connections to an unfamiliar destination were observed. These activities were treated as a simulated security incident requiring investigation.

7. Log Analysis

Time	Event ID	Description	Severity
09:42	4625	Multiple failed logins	Medium
09:44	4624	Successful login	High
09:46	4688	PowerShell process created	High
09:48	5156	Outbound network connection	High

8. Indicators of Compromise (IOCs)

- Repeated failed login attempts
- Unexpected successful login
- Suspicious PowerShell execution
- Unknown outbound connection
- Possible credential compromise
- Abnormal user activity

9. Incident Response and Containment

1. Verified security alerts.
2. Collected Windows logs.
3. Identified affected user account.
4. Isolated the affected system.
5. Blocked suspicious IP addresses.
6. Reset user credentials.
7. Performed antivirus and malware scans.
8. Preserved evidence and logs.
9. Restored normal operation.
10. Continued monitoring for recurring activity.

10. Detected Anomalies

- Consecutive authentication failures
- Suspicious process execution
- Unusual outbound traffic
- Elevated security alerts
- Abnormal login behavior

11. Results

The simulated investigation successfully identified potential Indicators of Compromise and demonstrated the importance of centralized log analysis. No permanent system damage was assumed because the incident was contained during the simulation.

12. Conclusion

This exercise demonstrated the basic incident response lifecycle from detection to containment and documentation. Log analysis and IOC identification are essential skills for cybersecurity professionals and help organizations reduce the impact of cyber threats.

13. Screenshot

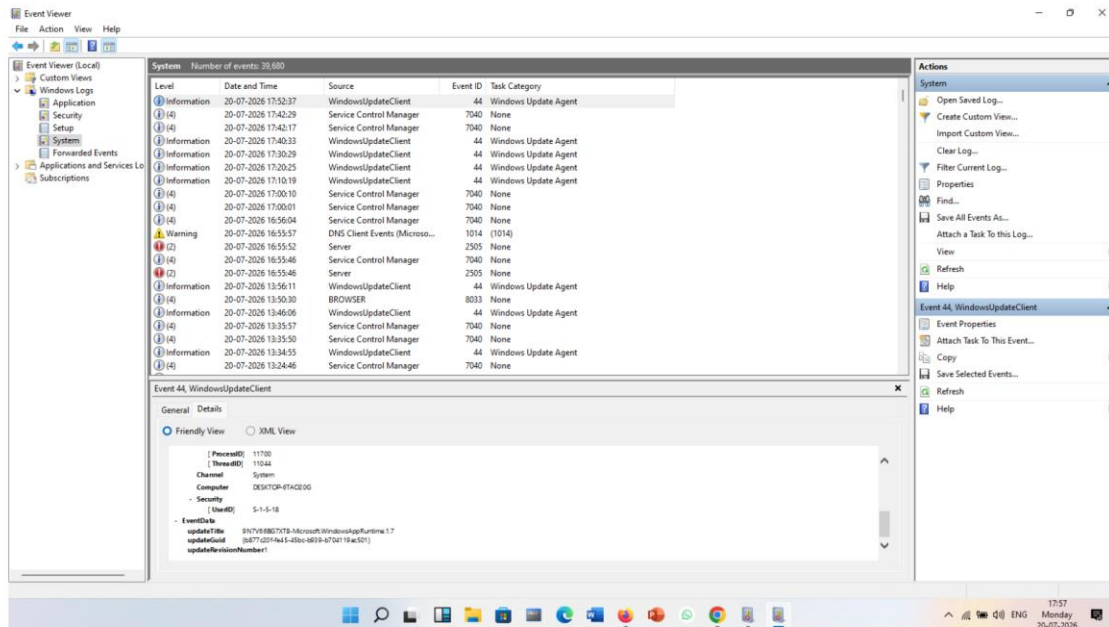


Figure 1: Windows Event Viewer / Security Log

Observation: This screenshot supports the incident response investigation by showing relevant log information used to identify suspicious activity and document the response process.

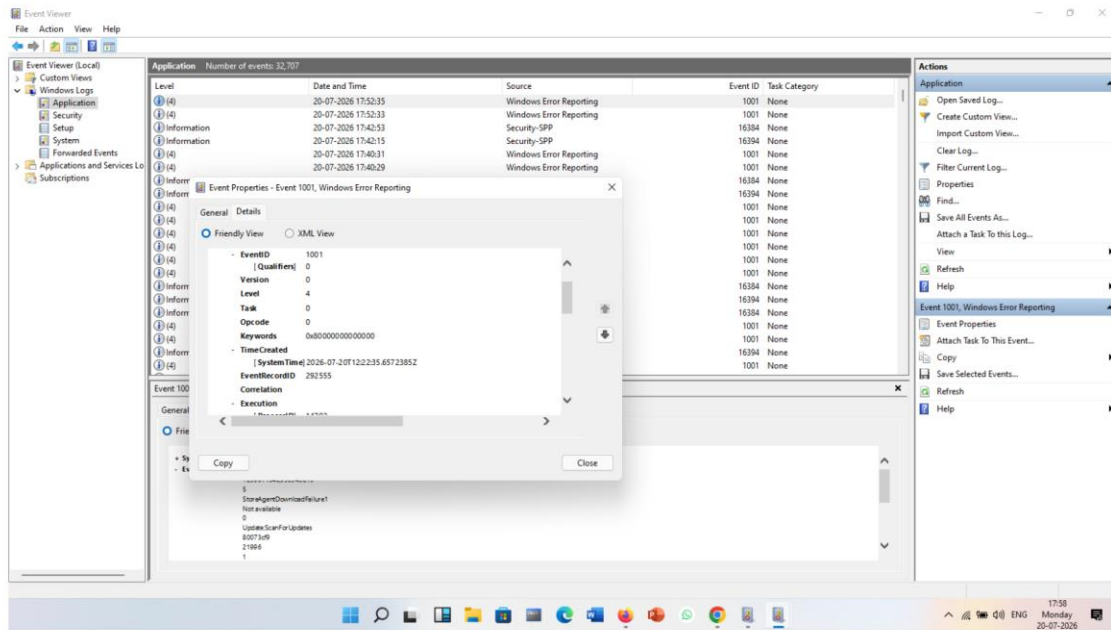


Figure 2: Suspicious Log Activity

Observation: This screenshot supports the incident response investigation by showing relevant log information used to identify suspicious activity and document the response process.

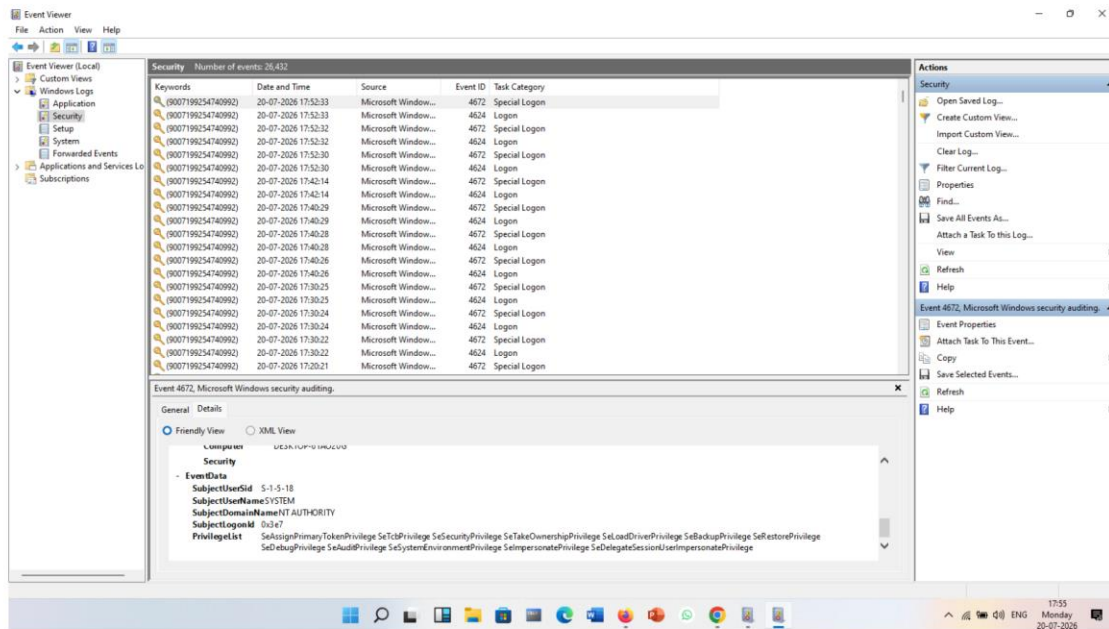


Figure 3: Indicators of Compromise

Observation: This screenshot supports the incident response investigation by showing relevant log information used to identify suspicious activity and document the response process.

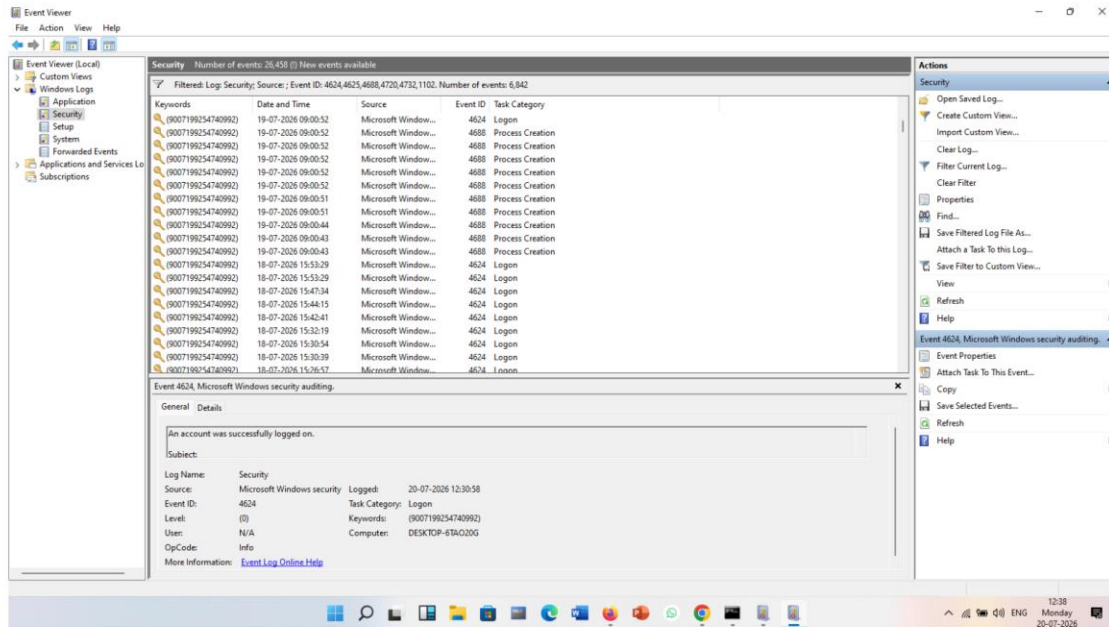


Figure 5: Response and Containment Evidence

Observation: This screenshot supports the incident response investigation by showing relevant log information used to identify suspicious activity and document the response process.